ETHIOPIA

COUNTRY REPORT ON GENOME EDITING LANDSCAPE ANALYSIS ON AGRICULTURE

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LIST OF ABBREVIATIONS/ACRONYMS

AAU	Addis Ababa University
ABNE	African Biosafety Network of Expertise
ABS	Access and Benefit Sharing
ACDI/VOCA	Agricultural Cooperative Development International / Volunteers in Overseas Cooperative Assistance
AGRA	Alliance for Green Revolution Africa
AOCC	African Orphan Crops Consortium
ARTP	Agricultural Research and Training Project
ATI	Agricultural Transformation Institute
AUDA-NEPAD	African Union Development Agency–New Partnership for Africa's Development
BETin	Bio and Emerging Technology Institute
BMGF	Bill and Melinda Gates Foundation
CBD	Convention on Biological Diversity
CFTs	Confined Field Trials
Codex	Codex Alimentarius Commission
CPB	Cartagena Protocol on Biosafety
CRISPR	Clustered Regularly Interspaced Short Palindromic Repeats
D&C	Deployment and Commercialization
DDPC	Donald Danforth Plant Science Center
EIAR	Ethiopian Institute of Agricultural Research
EIPA	Ethiopian Intellectual Property Authority
EFDA	Ethiopian Food and Drug Authority
EPA	Ethiopian Environmental Protection Authority
ESA	Ethiopian Seed Association
FAO	Food and Agriculture Organization
FDRE	Federal Democratic Republic of Ethiopia
GEd	Genome Editing
GETUP	Genome Editing in Tef for Uplifting Productivity
GTP	Growth and Transformation Plan
IGI	Innovative Genomics Institute
IP	Intellectual Property
IPRs	Intellectual Property Rights

ISAA	International Service for the Acquisition of Agri-biotech Applications
IITA	International Institute of Tropical Agriculture
LMOs	Living Modified Organisms
MEA	Multilateral Environmental Agreement
NBAC	National Biosafety Advisory Committee
NCA	National Competent Authority
NGO	Non-Governmental Organization
ODK	Open Data Kit
PIF	Policy and Investment Framework
PPP	Public-Private Partnership
R&D	Research and Development
RCBP	Rural Capacity Building Project
SIDA	Swedish International Development Cooperation Agency
SNV	Netherlands Development Organization
SSSfA	Striga Smart Sorghum for Africa
ToTs	Training of Trainers

EXECUTIVE SUMMARY

Background: Genome editing is increasingly seen as a promising technology that can boost productivity and output in Ethiopia's major agricultural sectors. With proper support, including sufficient funding and a supportive regulatory framework, GEd has the potential to solve many existing issues, such as crop and livestock diseases and low yields. Recognizing this, the Ethiopian government established the Environmental Protection Authority (EPA), an independent agency, to regulate this emerging technology.

The African Union Development Agency–New Partnership for Africa's Development (AUDA-NEPAD) commissioned a Genome Editing (GEd) Landscape Analysis for Ethiopia to assess the country's readiness to develop, deploy, and commercialize genome-edited products. Thus, Ethiopia's policy, regulatory, institutional, infrastructural, and technical capacities for modern biotechnology and GEd were evaluated, focusing on identifying key enablers and constraints across scientific, socioeconomic, cultural, and governance dimensions. Specifically, the analysis examined opportunities for GEd to address priority national needs at scale, including agricultural productivity, climate adaptation, food and nutrition security, post-harvest loss reduction, and diversification of healthy diets. It also identified staple and indigenous crops with high potential to improve livelihoods.

The assessment drew on both secondary data (published literature and institutional databases) and primary data collected through interviews, surveys, and questionnaires. Findings were synthesized across areas such as international treaty commitments, regulatory and policy frameworks, ongoing GEd projects and programs, human and institutional capacity, research and training infrastructure, priority crops and traits, intellectual property considerations, and funding mechanisms.

Key findings: Ethiopia has signed multiple international agreements and treaties related to food, climate change, and biodiversity. These include the Codex Alimentarius Commission (CAC), the United Nations Framework Convention on Climate Change (UNFCCC), the Convention on Biological Diversity (CBD), and the Cartagena Protocol on Biosafety (CPB). The country has incorporated these into its national food and environmental regulatory frameworks. Ethiopia's regulatory framework is fully functional with several biotech/GEd approvals. The EPA, an independent regulatory body established in 1995, regulates all activities related to Living Modified Organisms (LMOs) and Genetically Engineered (GE) products. The country enacted its biosafety law in 2009 (Proclamation No. 655/2009) and amended it in 2015. The Ethiopian Institute of Agricultural Research (EIAR) is one of the applicants that has secured several approvals to conduct research on transgenic crops under contained laboratory conditions, Confined Field Trials

(CFTs), and open-field trials (environmental clearance). Three GEd projects are ongoing in Ethiopia, focusing on essential food crops, sorghum for Striga resistance, teff for lodging resistance, and Ethiopian mustard oil with low erucic acid for improved quality. So far, the Teff lodging resistance GEd product has been approved for field trials.

Ethiopia considers socio-economic factors in its decision-making regarding biotech/GEd approval processes. These factors include increased productivity, nutrition, household income, youth employment, public perception, and gender equity. Knowledgeable and relevant individuals in GEd R&D were identified at key academic and national agricultural institutions. These institutions include the Bio and Emerging Technology Institute (BETin), Addis Ababa University, Haramaya University, and the Ethiopian Institute of Agricultural Research (EIAR). These institutions were found to have molecular laboratories with basic equipment and glasshouse infrastructure that can support R&D in GEd technology. However, some of these lab facilities lacked both funding and specialized equipment, such as high-throughput DNA sequencers. Key scientists from Ethiopia have participated in and benefited from GEd training at the IITA training lab in Nairobi, Kenya. Several staple, indigenous, and commercial crops were identified as currently under GEd and/or as candidates for improvement through GEd interventions in Ethiopia. These crops include Teff, sorghum, Ethiopian mustard oil, maize, wheat, and coffee.

The Ethiopian Intellectual Property Authority (EIPA) is responsible for administering and implementing intellectual property rights (IPRs) policies. Several private-sector companies could play a critical role in the commercialization of GEd products by providing testing sites and conducting on-farm trials of the GEd-developed lines. These include Green Agro Solutions PLC, MIDROC Investment Group, Luna Group, Corteva Agriscience, BASF (Germany), Tropic Biosciences (UK), and 2Blades Foundation. The funding agencies identified as supporting GEd projects include the United States Agency for International Development (USAID), the Gates Foundation, the Swedish International Development Cooperation Agency (SIDA), and the Ethiopian Government. A few key stakeholders with critical data and positive influence were also identified.

Policy Implications

Short term (next 12–24 months)

Ethiopia's new Guideline on Genome-Edited Products provides regulatory clarity and lays the foundation for confined field trials, particularly for crops such as teff. The immediate priorities include operationalizing the guidelines, implementing procedures, training reviewers, and developing transparent communication strategies to ensure predictable decision-making.

Policymakers must also strengthen biosafety capacity, establish clear labeling and traceability norms, and engage stakeholders (farmers, researchers, consumers, and civil society) to build public trust and social acceptance.

Medium to Long-term (2-5 years)

Ethiopia will need to integrate GEd into national agricultural and food security strategies, supported by policies that encourage local innovation, regional regulatory harmonization (aligned with AUDA-NEPAD), and market access planning, given divergent global trade rules. Strategic investments in CRISPR infrastructure, bioinformatics capabilities, and talent retention will be crucial, as will the development of stewardship frameworks to ensure equitable access to seeds and benefit-sharing. Over time, successful policy implementation could position Ethiopia as a regional leader in applying GEd to orphan crops such as teff and sorghum, contributing to climate resilience, productivity gains, and a comprehensive transformation of the food system.

Conclusion

Ethiopia showed "political goodwill" by replacing its former restrictive regulatory regime with a friendlier one to facilitate the adoption of modern biotechnology. As a result, Ethiopia has field-tested and released some biotech products for commercial cultivation, including Bt cotton and TELA maize. More recently, Ethiopia's regulatory system has been strengthened by the publication of the Guideline on the Regulation of Genome-Edited Products. This has enabled researchers to engage in GEd R&D, making Ethiopia one of the few African countries to adopt this technology. This has paved the way for three active GEd projects on key food-security crops and traits to mitigate challenges posed by climate change. However, the funding landscape for GEd R&D is driven by external investors, resulting in a substantial funding gap and limited access to specialized equipment.

1. INTRODUCTION

1.1. Description of Ethiopia's Agricultural Landscape

Ethiopia's agricultural landscape is dominated by smallholder farming, which provides livelihoods for more than 80% of the population and remains the backbone of the national economy. Agriculture contributes about 35% of Ethiopia's GDP and generates 90% of export revenue (Yigezu, 2021). The sector is mainly rain-fed and subsistence-oriented, with farmers cultivating an average of less than two hectares of land. Key staple crops include teff (*Eragrostis tef*), maize (*Zea mays*), sorghum (*Sorghum bicolor*), wheat (*Triticum spp*.), and barley (*Hordeum vulgare*), alongside pulses and oilseeds. At the same time, coffee (*Coffea arabica*) is the country's main export crop. Livestock production is also significant, with Ethiopia hosting one of Africa's largest cattle populations, contributing to food, income, and draft power. Despite its importance, agriculture continues to face persistent challenges, including low productivity, land degradation, limited irrigation, and vulnerability to climate variability, particularly drought. In recent years, government strategies and investment plans have focused on modernizing agriculture through improved inputs, expanded irrigation, climate-smart practices, and the integration of technologies such as biotechnology and GEd to boost resilience and productivity.

1.2 National Frameworks and Investments

Genome editing in the country is increasingly being incorporated into broader national strategies, aligning with Ethiopia's Agricultural Development and Investment Plans. These plans focus on food security, climate resilience, and productivity improvements for key staple crops such as teff, sorghum, and wheat. The country's biotechnology and biosafety regulatory system is based on the amended 2015 Biosafety Proclamation. The Guideline on the Regulation of Genome-Edited Products supports national science, technology, and innovation efforts by building local capacity for advanced breeding methods and linking to climate adaptation and nutrition initiatives that promote resilient crop varieties and better dietary quality. The framework positions GEd as a cross-cutting tool embedded in Ethiopia's development priorities, with the potential to accelerate agricultural transformation, attract investment, and enhance the country's role as a regional leader in Agri-biotech.

Ethiopia has actively aligned its biotechnology and GEd governance with regional frameworks, especially those led by the AUDA-NEPAD and the African Biosafety Network of Expertise (ABNE). The country has taken part in regional consultations, training sessions, and efforts to harmonize policies that prioritize science-based, proportionate regulation of modern biotechnologies. Ethiopia's adoption of the Guideline on Regulation of Genome-Edited Products illustrates this regional influence, using AUDA-NEPAD guidance to differentiate gene-edited products from transgenic GMOs. By engaging with these continental platforms, Ethiopia enhances its domestic capacity, promotes policy consistency, and positions itself to support regional trade, knowledge exchange, and regulatory convergence, especially for genome-edited crops such as teff and sorghum, aligning with Africa's broader agricultural innovation goals.

1.3. Engagement in Regional Frameworks

Ethiopia is proactively aligning its regulations on genome editing and modern biotechnology with both national laws and regional Economic Communities (RECs). As a member of the Common Market for Eastern and Southern Africa (COMESA), Ethiopia is part of initiatives such as the 2016 Biosafety & Biotechnology Implementation Plan (COMBIP), which aims to standardize biosafety procedures, create risk assessment databases, and map genome-editing advancements among member states. Additionally, in 2020, Ethiopia joined multiple African countries in regional discussions on biotechnology regulation through the ABNE and AUDA-NEPAD.

1.4. Challenges and Opportunities: Justification of GEd Assessment based on Current Issues Identified in Ethiopia

Ethiopia's need to evaluate and implement GEd technologies is driven by an urgent need to address critical agricultural and food system challenges, while also recognizing considerable opportunities for transformative change. This landscape analysis is warranted by clearly documented issues in Ethiopian agriculture that traditional methods have failed to resolve effectively (Abate et al., 2015; Diro et al., 2019). The combination of these ongoing challenges with emerging scientific advances makes a strong case for thorough GEd assessment and strategic resource allocation.

1.4.1. Challenges Justifying GEd Assessment

Agricultural Productivity and Food Security Crisis: Ethiopia faces persistent food insecurity affecting millions annually, driven by chronically low agricultural productivity (World Bank, 2020). Average cereal yields remain below global and regional benchmarks. Teff yields average approximately 1.5 tons per hectare compared to potential yields of more than 3 tons (Assefa et al., 2015; Tadele, 2019), while sorghum and maize productivity similarly lag (Abate et al., 2015). Conventional breeding programs, though ongoing, have not delivered sufficient genetic gains to keep pace with population growth and changing dietary demands (Dawson et al., 2021). This productivity gap directly justifies assessing GEd as a precision tool to accelerate varietal improvement for traits such as yield potential, drought tolerance, and nutrient use efficiency that have proven intractable with traditional methods (Varshney et al., 2021).

Climate Vulnerability and Environmental Degradation: Recurrent drought, increasingly erratic rainfall patterns, and soil degradation threaten the livelihoods of the 80% of Ethiopians dependent on rain-fed agriculture (World Bank, 2020; FAO, 2021). Climate models project increased aridity in key agricultural zones, necessitating rapid development of climate-resilient crop varieties (Conway & Schipper, 2011; Adem *et al.,* 2022). Conventional breeding cycles of 8–12 years are ill-suited to the urgency of climate adaptation (Tester & Langridge, 2010). GEd assessment is justified by the technology's demonstrated capacity to shorten breeding timelines and introduce precise stress-tolerance traits that could enhance resilience in orphan crops such as teff and sorghum, which lack the intensive research investment directed toward global commodities (Tadele, 2019; Gao, 2021).

Biotic Stresses and Pest/Disease Pressures: Crop losses from pests and diseases remain substantial, with emerging threats such as fall armyworm (*Spodoptera frugiperda*) and wheat rusts causing significant damage (CABI, 2020; FAO, 2021). The parasitic weed *Striga* (*Striga hermonthica*) infests approximately 50% of Ethiopian cereal cropland, causing annual losses exceeding US$100 million (Gressel, 2018; Runo & Kuria, 2019). Chemical and cultural control methods face practical limitations, cost barriers, and environmental concerns (Gressel, 2018). The ongoing Striga Smart Sorghum for Africa project exemplifies how GEd assessment can target specific, high-impact biotic constraints that have resisted integrated pest management approaches (Iraki et al., 2023), thereby justifying expanded assessment of similar applications.

Institutional and Infrastructure Constraints: Limited laboratory infrastructure, dependence on external technical collaborations, and critical shortages of trained personnel in advanced molecular biology constrain Ethiopia's innovation capacity (Karembu et al., 2023). Fragmented seed systems and weak extension services further impede the delivery of technology from research stations to farmers (Spielman et al., 2013; Abate et al., 2015). Assessment of GEd readiness is essential to identify specific infrastructure gaps, training needs, and institutional strengthening requirements that must precede effective technology deployment (AUDA-NEPAD, 2022).

Socio-Political and Market Factors: Regional political instability has disrupted agricultural activities, diverted government priorities, and strained institutional capacity for research oversight (FAO, 2021; World Bank, 2022). Public understanding of biotechnology remains limited, with misinformation creating barriers to social acceptance (Glover et al., 2016; Adem et al., 2022). These factors necessitate a comprehensive assessment not merely of technical capabilities but also of governance frameworks, stakeholder engagement mechanisms, and communication strategies essential for responsible innovation (Hartley et al., 2022).

1.4.2. Opportunities Supporting GEd Investment

Enabling Policy Environment: Ethiopia's commitment to agricultural transformation through successive National Agricultural Development and Investment Plans provides strong policy alignment for GEd assessment (FDRE, 2019, 2021). The 2025 Guideline on Regulation of Genome-Edited Products establishes unprecedented regulatory clarity, distinguishing gene-edited products from transgenic GMOs and creating specific pathways for research and commercialization (EPA, 2025; AfriSCI Dialogue, 2024). This policy progress justifies immediate assessment of implementation readiness and capacity needs.

Strategic Partnerships and Capacity Building: Active collaborations with international institutions, including the Donald Danforth Plant Science Center, Michigan State University, and the African Orphan Crops Consortium, demonstrate viable pathways for technology access and human capital development (Beyene et al., 2022; Mba et al., 2022). The three ongoing GEd projects (teff lodging resistance, Ethiopian mustard oil quality, Striga-resistant sorghum) provide foundational experience justifying expanded assessment of pipeline opportunities and scaling potential (Ngure & Karembu, 2023; Iraki et al., 2023).

Orphan Crop Leadership: Ethiopia's unique position as the center of origin and diversity for teff, along with significant diversity in sorghum and finger millet, creates a competitive advantage in developing GEd applications for under-researched crops with limited global private-sector investment (Assefa et al., 2015; Cheng et al., 2018). Assessment of GEd opportunities in these crops addresses both national food security and potential regional market leadership (Tadele, 2019; Mba *et al.,* 2022).

Economic Transformation Potential: Successful GEd deployment could reduce dependence on imports for key commodities, enhance agricultural export competitiveness, and generate rural employment through improved productivity (Admassie et al., 2017; Tadele, 2019). Assessment of economic returns and value chain impacts is essential for justifying public investment and attracting private-sector engagement (Zilberman et al., 2018; Entine et al., 2021).

The convergence of these challenges and opportunities creates an imperative for rigorous, evidence-based assessment of Ethiopia's readiness to harness genome editing (Abate et al., 2015; AUDA-NEPAD, 2022). This landscape analysis accordingly examines regulatory frameworks, institutional capacities, infrastructure requirements, human capital needs, and stakeholder ecosystems not as abstract inventories, but as functional prerequisites for addressing the specific agricultural constraints identified above. The analysis seeks to determine whether and how GEd can be effectively directed toward Ethiopia's most pressing food system challenges, ensuring that technology assessment is grounded in national development priorities rather than technology-driven imperatives.

1.5. Objectives of the Landscape Analysis

The general (overall) objective of this Genome Editing (GEd) Landscape Analysis is to provide an in-depth assessment of existing policies, infrastructure, institutional arrangements, and technical capabilities related to product development and commercialization in Ethiopia.

Specifically, the objectives of this Landscape Analysis were to:

i. Provide an evidence-based description and analysis of the status of modern biotechnology and GEd in Ethiopia, highlighting key trends, intervening factors, and areas requiring attention. This includes examining fundamental aspects such as scientific/technical capacity, political and geopolitical contexts, social dynamics, human capital, cultural traditions, and other factors that support or hinder advances in the application of genome editing in agriculture and food systems.
ii. Identify emerging needs in Ethiopia that GEd can readily address, particularly those requiring rapid responses at scale. These needs focus on food systems, including enhancing agricultural productivity, reducing post-harvest losses, adapting to climate change, ensuring food and nutrition security, and promoting diversified and healthy diets.
iii. Identify staple and indigenous crops based on Ethiopia's national context that can improve livelihoods through enhanced food security, better nutrition, climate resilience, and sustainable productivity.

2. APPROACH / METHODOLOGY

Data were collected from both secondary and primary sources. Secondary data were gathered from published literature and institutional databases, including official reports, peer-reviewed publications, and relevant organizational websites. Primary data were collected through live interviews using the Online Data Collection Kit (ODK), structured surveys, and email-based questionnaires shared in Microsoft Word format. Primary data collection served to validate, supplement, and strengthen the secondary data. Data from both sources were subsequently analyzed, synthesized, and packaged under the following thematic areas.

2.1. Regulatory and Policy Frameworks for Biotechnology and Genome Editing

Components of the biotechnology and GEd regulatory frameworks were identified and documented using secondary sources (published literature and institutional websites) and primary data from interviews, surveys, and email correspondence.

2.2. Genome Editing Projects

Information on biotechnology and GEd projects was compiled from primary and secondary data sources. This included documentation of target crops, livestock, fisheries, and forestry species; traits under development; key stakeholders and partnerships involved; and funding sources. The synthesized data were used to:

i. identify emerging economic, social, and environmental/climate-related needs; and
ii. assess the status of human capacity and infrastructure supporting genome editing technologies in the target countries.

2.3. Traditional and Staple Crops for Genome Editing Improvement

Data on GEd projects were further disaggregated to identify staple, indigenous, and commercial crops with the highest potential for socio-economic impact. Selection criteria included national acceptance, resource requirements, scalability, and likelihood of successful development and deployment.

2.4. Institutional Capacity (Human Capital, Infrastructure and Equipment)

Data on institutional capacity, including human resources, laboratory and field infrastructure, and equipment for genome editing R&D, commercialization, and scaling, were collected from respondents during the primary data collection exercise.

2.5. Stakeholder Mapping

Key stakeholders and institutions, and, where appropriate, individual experts, were identified to provide critical information on existing biotechnology and genome-editing interventions. Stakeholders were categorized into five (5) groups, as reflected in the data collection tools:

- Regulatory agencies
- Research organizations and institutions
- Universities
- Private sector and industry
- Government ministries, agencies, and policymakers

2.6. Database Systems and Data Management

Customized data collection tools (questionnaires) and digital platforms were developed to support primary data collection. Questionnaires were tailored to each stakeholder category and used to generate structured datasets assessing Ethiopia's preparedness and capacity to adopt, implement, and scale genome-editing technologies.

Enumerators were identified, recruited, and trained through an online induction program on the use of ODK and associated data-collection tools. Training was conducted by the IT team and train-the-trainers (ToT) from the consortium. All tools and platforms were pre-tested prior to deployment. Both methodological and data triangulation approaches were employed to reduce bias and enhance the validity and reliability of findings. All collected data were encrypted and stored on a secure, centralized server to ensure participant confidentiality and compliance with ethical standards.

2.7. Data Synthesis and Data Analysis

Data collected was synthesized and packaged into appropriate information, and, where applicable, tables were generated.

Variables investigated included, but were not limited to:

i. signatory to multinational food and environmental agreements;
ii. regulatory framework components;
iii. applications received and approved (field testing, registration, and commercialization);
iv. R&D institutions, personnel, and laboratory/field facilities;
v. projects;
vi. crops and livestock species; and
vii. traits under GEd development.

2.8. Interactive Map

An interactive map, comparable to the Agenda 2063 Dashboard, was developed to visualize genome editing activities and capacities across Africa and is accessible at:
https://genome.africa/

3. RESULTS

3.1. Membership in Multilateral Environment Treaties and Agreements

Ethiopia has established a comprehensive legal and institutional framework to guide biotechnology and genome editing, anchored in commitments under multilateral environmental agreements (*See Table 1*). The country is a Party to the Convention on Biological Diversity (CBD). It has ratified its major instruments, including the Cartagena Protocol on Biosafety (CPB), which provides the foundational framework for regulating living modified organisms (LMOs) (Secretariat of the Convention on Biological Diversity, 2000; EPA, 2009). Ethiopia has also acceded to the Nagoya Protocol on Access and Benefit Sharing, reinforcing sovereign rights over genetic resources and equitable benefit-sharing with provider communities (FAOLEX, 2012).

Table1. Status of Country Participation in Key Multilateral Environmental Agreements (MEAs)

Multilateral Environmental Agreements (MEAs) / Treaties	**Year (Signed/Ratified)**	**Reference**
Codex Alimentarius Commission is a joint body of the Food and Agriculture Organization (FAO) and the World Health Organization established to develop international food standards, guidelines, and codes of practice. critical for the risk assessment of food produced through genome editing	Member 1968	https://www.fao.org/fao-who-codexalimentarius/about-codex/members/en/
Convention on Biological Diversity	Signed: 1992 Ratified: 2003	https://www.cbd.int/doc/legal/cbd-en.pdf
Cartagena Protocol on Biosafety	Signed: 2000 Ratified: 2003 Enforcement: 2004	https://bch.epa.gov.et/?utm
United Nations Framework Convention on Climate Change (UNFCCC)	Ratified 1994	https://unfccc.int/sites/default/files/resource/Ethiopia_First%20BUR.pdf?utm_s
Nagoya Protocol on Access and Benefit Sharing (ABS)	Proclamation 2012 (No. 753/2012)	http://faolex.fao.org/docs/pdf/eth160766.pdf

3.2. Policies that Advance or Hinder Adoption of Modern Biotech and GEd

Ethiopia is currently transitioning to modern agricultural biotechnology, moving from highly cautious policies to a more supportive environment. Key policies promoting the adoption of modern biotechnology and GEd include:

National Biotechnology Policy and Strategy (2002): The plan proposes developing and expanding biotechnology education and R&D to enhance national capabilities in the life sciences. It provides a broad framework for building biotechnology capacity across agriculture and other sectors.

National Science, Technology, and Innovation Policy (2006): This policy, formulated by the Ethiopian Science and Technology Agency, aims to build national capability to generate, select, import, and apply technologies for socio-economic development (Sube et al., 2025).

EIAR Agricultural Biotechnology Research Strategy (2016-2030). The strategy guides EIAR in prioritizing and implementing agricultural research to deliver technologies, information, and knowledge that will strengthen the agricultural sector. The strategy emphasizes adhering to biosafety regulations while advancing biotechnology programs in collaboration with various government universities.

3.3. National Regulatory Framework

3.3.1. Biosafety Act/Law, Biosafety Regulations/Decrees, and GEd Guidelines

At the national level, the Biosafety Proclamation (Amendment) No. 896/2015 is the main legal framework governing biosafety. It replaces the more restrictive Proclamation No. 655/2009, which limited research and innovation by focusing exclusively on conservation and not considering the benefits of biotechnology (*see Table 2*). In 2025, this legislative framework was further enhanced by the publication of the Guideline on the Regulation of Genome-Edited Products, which establishes a risk-based, case-by-case assessment process for gene-edited organisms that is different from GMOs (EPA, 2025; AfriSCI Dialogue, 2024).

The EPA serves as the national competent authority (NCA) for biosafety, mandated to grant permits, conduct risk assessments, and oversee compliance throughout the GEd product development continuum (EPA, 2015). The Bio and Emerging Technology Institute (BETin) supports policy coordination, research prioritization, and public communication functions. At the same time, the EIAR and universities constitute the core research and capacity-building infrastructure. The Ethiopian Food and Drug Authority (EFDA) is responsible for food and feed safety assessment, ensuring that genome-edited products entering the food system meet established safety standards. Ethiopia's membership in the Codex Alimentarius Commission further aligns national food safety assessment with international norms, critical for both domestic consumer protection and future trade facilitation (FAO/WHO, 2023).

The National Biosafety Advisory Committee (NBAC) represents a critical coordinating mechanism, comprising 15 institutions with membership appointed by the Prime Minister to ensure high-level government commitment (EPA, 2015). Participating institutions span ministerial portfolios including agriculture, health, trade and industry, justice, and planning, alongside technical agencies such as the Ethiopian Biotechnology Institute, Ethiopian Biodiversity Institute, Ethiopian Standards Agency, and EIAR; academic institutions including Addis Ababa University Institute of Biotechnology and Addis Ababa Science and Technology University; and civil society representatives from consumer associations and non-governmental organizations. This diverse composition ensures multidisciplinary input into biosafety decision-making while embedding stakeholder perspectives in governance processes (AUDA-NEPAD, 2022). The regulatory approval process is presented in Figure 1.

The GEd value chain in Ethiopia spans six interconnected stages: (1) Research and Discovery, (2) Product Development, (3) Regulatory Assessment, (4) Seed Production and Distribution, (5) Commercial Cultivation, and (6) Market Access and Trade. The following analysis identifies existing key activities, lead institutions, and the regulatory framework. A summary of the regulatory and institutional landscape for GEd in Ethiopia is presented in Table 3.

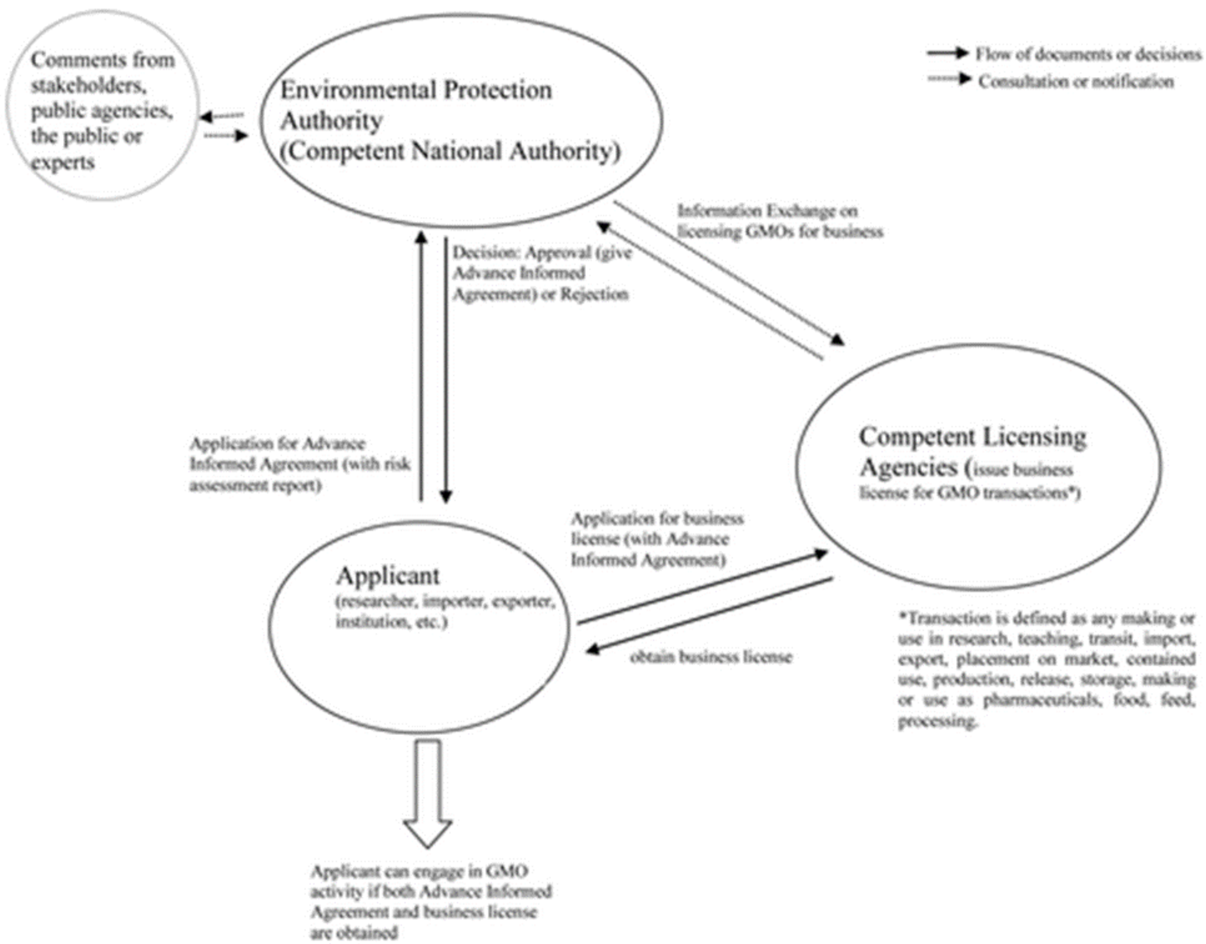

Figure 1. Regulatory framework for approval of all GMO activities in Ethiopia; Source: Abraham A. (2013).

Table 2. Regulatory and Institutional Landscape for Genome Editing (GEd) in Ethiopia

Institution	Mandate / Relevance to GEd	Regulatory instruments	Year of enactment/ publication	Coverage/ scope (not related to the EPA role)	Reference
Environmental Protection Authority (EPA)	Developing and implementing the legal and regulatory framework for genome editing and modern biotechnology in Ethiopia.	Biosafety Proclamation No.655/2009 National Biotechnology Law Biosafety (Amendment) Proclamation No. 896/2015	2009 2015	R&D, Commercialisation, Trade, etc.	https://bch.epa.gov.et/downloads-3/
		National Biosafety (Implementation, etc.) Regulations 2017	2017	R&D, Commercialisation, Trade, etc.	https://bch.epa.gov.et/downloads-3/
		Directive No. 05/2018 (To provide Risk Assessment of Genetically Modified Organisms (GMOs)	2018	R&D, Commercialisation, Trade, etc.	https://bch.epa.gov.et/downloads-3/
		Guidelines on stacked genes	2024	R&D, Commercialisation, Trade, etc.	https://bch.epa.gov.et/downloads-3/
		Guideline for the Regulation of Genome-Edited Plants (Animals, Plants, and Microbes) in Ethiopia	2024	R&D, Commercialisation, Trade, etc.	https://webinar.afriscidialogue.org/ethiopia-publishes-guidelines-on-regulation-of-genome-edited-products/?utm

Table 3. Institutional and Regulatory Landscape of GEd

Stage	Key Activities	Lead Institutions	Regulatory Framework
1. Research & discovery	• Basic research on target traits • Gene identification and validation	• Addis Ababa University (AAU) • Ethiopian Institute of Agricultural Research (EIAR) • Bio and Emerging Technology Institute	• Biosafety Proclamation 655/2009 (as amended 896/2015) • 2025 Guideline on Regulation of Genome-Edited Products

	• CRISPR guide RNA design • Bioinformatics and genomic analysis • Proof-of-concept experiments	(BETin) • Haramaya University • International partners (Danforth Center, MSU)	• Institutional biosafety committees • Laboratory containment standards (BSL1/BSL2)
2. Product development	• Transformation and regeneration • Molecular characterization • Greenhouse evaluation • Confined field trials (CFTs) • Agronomic performance testing • Environmental risk assessment	• EIAR (lead for crops) • AAU (academic research) • EPA (oversight) • Regional agricultural research institutes • Private sector (seed companies)	• 2025 Guideline on Regulation of Genome-Edited Products • Directive on CFTs (under development) • National Biosafety Advisory Committee (NBAC) oversight • Environmental release protocols
3. Regulatory assessment	• Application review and approval • Risk assessment (food, feed, environmental) • Decision-making and authorization • post-release monitoring • Compliance enforcement	• EPA (National Competent Authority) • Ethiopian Food and Drug Authority (EFDA) (food/feed safety) • NBAC (advisory) • BETin (technical support) • Ministry of Justice (legal)	• Biosafety Proclamation 896/2015 • 2025 Guideline on Regulation of Genome-Edited Products • Cartagena Protocol on Biosafety (CPB) • Codex Alimentarius standards • Nagoya Protocol on ABS
4. Seed production & distribution	• Variety registration and release • Breeder seed production • Foundation and certified	• Ministry of Agriculture (variety release) • EIAR (breeder seed) • Ethiopian Seed Enterprise • Private seed companies (MIDROC,	• Seed Proclamation and regulations • Variety release procedures • Seed certification standards • Quality control protocols

	seed multiplication • Quality assurance and certification • Distribution to farmers	Luna, Green Agro) • Regional seed enterprises • Farmer cooperatives	• (GEd-specific standards under development)
5. Commercial cultivation	• Farmer adoption and cultivation • Agronomic extension and support • Input supply (fertilizer, pest management) • Harvest and post-harvest handling • On-farm stewardship	• Ministry of Agriculture (extension) • Regional agricultural bureaus • Agricultural Transformation Agency • Farmer cooperatives • Input suppliers • NGOs and development projects	• Agricultural extension policies • Input supply regulations • Pesticide management laws • Stewardship guidelines (under development) • Farmer protection regulations
6. Market access & trade	• Domestic market development • Regional and international trade • Consumer acceptance and labeling • Value chain integration • Benefit sharing and royalty management	• Ministry of Trade and Regional Integration • Ethiopian Standards Agency • Customs Commission • Regional economic communities (EAC, IGAD, AU) • Commodity exchanges • Exporters and agribusiness	• Trade and commercial laws • Biosafety and food safety regulations • Regional trade agreements • International standards (Codex, IPPC) • Nagoya Protocol on ABS • Seed trade regulations

3.4. Socio-economic considerations for decision-making

Ethiopia has integrated socio-economic considerations into regulatory decision-making processes for biotechnology and GEd products, particularly during environmental release approvals and market placement. Such considerations are incorporated through public participation mechanisms and play a critical role in determining whether biotechnology and GEd products are advanced or constrained in adoption. These considerations encompass economic, social, cultural, human, and environmental dimensions, including costs and market opportunities, employment and livelihoods, equity, consumer choice, food security, and impacts on biodiversity and indigenous knowledge systems as elaborated below:

Economic impacts include:

Market opportunities and employment: Given Ethiopia's predominantly smallholder farming system, the emphasis should be on genome-edited traits that boost yield stability, reduce input needs, and perform reliably under low-input and climate-stressed conditions. Benefits should be evaluated not just for productivity but also for effects on household income, nutrition, and youth employment. While GEd could create new market opportunities, critics worry about market displacement and effects on traditional products. Market readiness and seed system capacity are critical determinants of impact; without effective commercialization and extension pathways, technological gains may not translate into improvements in livelihoods.

Input Costs: Another key socio-economic consideration is the affordability of GEd-derived inputs, such as seeds, particularly for resource-poor farmers.

Social and cultural impacts include:

Equity and equality: Ensuring fair distribution of benefits and costs across different population groups, including gender and age considerations. Thus, embedding socioeconomic criteria will ensure genome editing delivers equitable, sustainable, and nationally relevant outcomes.

Consumer choice: Access to information and consumers' ability to make informed choices based on facts would also be a key consideration in regulatory decisions regarding GEd products.

Cultural practices: The decision would also be informed by the potential impacts of GEd technology on traditional seed systems, indigenous knowledge, and local customs, which require careful evaluation to prevent the erosion of cultural heritage.

Food security: Assessment of how biotech/GEd products influence food availability, access, and stability is a key determinant of decision-making in the country.

Public perception: Public trust in GEd products would also play a decisive role, underscoring the need for transparent decision-making and proactive risk communication to differentiate GEd technology from GMOs.

Environmental considerations include:

Impacts on biodiversity and traditional crops: Potential displacement of traditional varieties and loss of genetic diversity, with direct socio-economic implications, are fundamental considerations in the approval of any technology for commercial release.

Impacts on indigenous technologies: Concerns that replacing traditional farming methods could disrupt local agricultural systems and associated cultural practices have also been a major socio-economic consideration in the deregulation of GMOs, and might be so for GEd.

3.5. Factors that Hinder Application of Modern Biotech and GEd

The application of modern biotechnology and Genome Editing (GEd) in Ethiopia is shaped by a complex interaction of political, geopolitical, social, human, cultural, and traditional factors. Some of these factors strongly support the adoption of biotechnology, while others slow research, regulation, public acceptance, and commercialization.

Political Factors: Ethiopia has created a complex environment for the adoption of modern biotechnology, transitioning from a historically cautious, restrictive stance to a more permissive, yet still contested, approach. While the government has shown increased political will to adopt biotechnology for food security and industrial growth by 2026, several political, regulatory, and social factors continue to hinder widespread adoption. Key political factors hindering the adoption of modern biotechnology in Ethiopia include:

- ***Intense Political and Civil Society Opposition**:* A coalition of civil society organizations, environmental activists, and consumer associations that actively campaign against GMOs, citing risks to biodiversity, food sovereignty, and potential economic dependence on foreign multinational seed companies.
- ***Influence of Anti-Biotech Lobbyists on Policy Makers*:** Lobbying against GMOs with targeted misinformation that has created skepticism among some policymakers and officials, challenging the government's ability to make swift, science-based decisions on biotechnology.

- ***Political Prioritization of Organic Agriculture:*** Political actors, some of which promote Ethiopia as a "non-GMO" or organic-only agricultural exporter, create a direct policy clash with advocates of modern biotechnology.

Geopolitical Factors: Ethiopia works with international and regional groups like AUDA-NEPAD and ABNE to build biosafety capacity and promote harmonized biotech regulation. Although the country has recently become more receptive to biotechnology and genome editing, geopolitical factors still hinder progress.

- ***Dependence on Foreign Donors and External Funding:*** Biotechnology research in Ethiopia relies heavily on international donors (e.g., USAID, DDPSC, Corteva Agriscience) and multinational research organizations. This dependence creates geopolitical vulnerability because national biotechnology priorities may become influenced by donor agendas, foreign intellectual property systems, and external research interests. It also limits Ethiopia's autonomy in setting independent long-term biotechnology strategies.
- ***Regional and Continental Regulatory Fragmentation:*** Africa lacks harmonized biotechnology regulations, and countries differ widely in the application of modern biotech and genome editing. Lack of harmonized African biosafety systems complicates commercialization and discourages investment in biotechnology innovation (Rabuma *et al.,* 2024).

Social and Human Factors: Several social and human factors continue to hinder the adoption of modern biotechnology and genome editing in Ethiopia. These factors influence public acceptance, policy implementation, R&D, and farmer-level utilization of biotechnology innovations, including low public awareness and low scientific literacy. Limited public understanding of biotechnology and genome editing technologies, such as CRISPR, has slowed adoption. Studies in Ethiopia show that acceptance of biotechnology increases with higher education and scientific exposure (Sbhatu, 2010).

- ***Weak Extension and Technology Dissemination Systems:*** Inadequate agricultural extension services and farmer outreach have greatly slowed the adoption of biotechnology. Weak linkages between research institutions, private companies, and farmers further hinder dissemination.

Culture and Tradition Factors: Ethiopia has a long tradition of agricultural innovation and farmer adaptation to harsh environmental conditions. Culture and tradition can either slow or facilitate acceptance depending on how technologies are introduced, communicated, and aligned with local values and farming systems. Culture and traditions include:

- ***Cultural Importance of Indigenous Crops:*** Ethiopia is a center of origin and diversity for crops such as teff, enset, barley, sorghum, and coffee. These crops are not only economic commodities but also carry strong cultural and symbolic value. Communities may therefore be cautious about genetic modification or genome editing of culturally significant crops because of fears that the technology could alter taste, quality, traditional uses, or cultural authenticity (Borrell *et al.,* 2021).
- ***Fear of Losing Seed Saving Culture:*** Traditional practices involve saving, exchanging, and reusing seeds. The introduction of proprietary, patented biotechnological seeds creates a perceived threat to this, as farmers are prohibited from saving seeds and become dependent on annual purchases. In addition, concerns about ownership of genetic resources are particularly sensitive in Ethiopia because of the country's rich biodiversity and history of protecting indigenous genetic resources.

3.6. Genome Editing Programs and Projects

Ethiopia currently maintains three active genome editing projects focusing on important food crops: Ethiopian mustard (*Brassica carinata*), teff (*Eragrostis tef*), and sorghum (*Sorghum bicolor*) (Ngure & Karembu, 2023). All projects target crop species and exclude livestock, fisheries, marine resources, or agroforestry applications. Two projects remain in research and development phases, while one has advanced to deployment and commercialization readiness *(see Table 4*).
The Ethiopian mustard oil quality improvement project, hosted at the Institute of Biotechnology, Addis Ababa University, aims to reduce erucic acid content in mustard oil using CRISPR-Cas9 technology. Funded by the Swedish International Development Cooperation Agency (SIDA) and implemented in collaboration with international partners, this project addresses the emerging potential of oilseeds for local edible oil production and industrial applications. The research and development orientation reflects both the technical complexity and the market development requirements of novel varieties.
The Genome Editing in Tef for Uplifting Productivity (GETUP) project represents Ethiopia's most advanced GEd initiative, having progressed to deployment and commercialization readiness (Beyene et al., 2022). This collaborative effort, involving Michigan State University, the Ethiopian Institute of Agricultural Research, Corteva Agriscience, and the Donald Danforth Plant Science Center, uses CRISPR-Cas9 to develop lodging-resistant semi-dwarf varieties. With $4.9 million in funding from the Gates Foundation, the project has successfully generated tetra-allelic mutations in the SEMIDWARF-1 (SD-1) gene, demonstrating proof of concept for improved agronomic performance (Beyene et al., 2022).

The Striga Smart Sorghum for Africa Project, led by Kenyatta University with Ethiopian participation through Addis Ababa University, aims to develop sorghum resistant to parasitic weeds using CRISPR-Cas9-mediated genome editing (Iraki et al., 2023). Funded by the United States Agency for International Development (USAID) at $3.5 million over five years, with implementation support from the International Service for the Acquisition of Agri-biotech Applications (ISAAA) AfriCenter, this project addresses one of the most devastating constraints on cereal production in sub-Saharan Africa.

The collaborative structure exemplifies regional research integration while building Ethiopian scientific capacity through technology transfer and training. Institutional analysis reveals Addis Ababa University as the pivotal domestic institution, hosting two of three ongoing projects and maintaining the most advanced molecular biotechnology infrastructure. The Ethiopian Institute of Agricultural Research serves as the essential bridge between research and application. At the same time, international partnerships provide access to proprietary technologies, technical training, and regulatory experience, thereby accelerating the development of domestic capabilities.

Table 4. Genome Editing Projects and Programs in Ethiopia

Projects/ Programs (organism)	**Trait**	**Collaborating partners**	**GEd Technique**	**Stage (Lab, field trial, commercialization)**	**Funding million (US$)**	**Funding source**	**Reference**
Improving the oil qualities of Ethiopian mustard (*Brassica carinata*)	Low erucic acid	Institute of Biotechnology, Addis Ababa University	CRISPR-Cas9	Research and Development (R & D)	Not specified	Swedish International Development Cooperation Agency (SIDA)	Ngure & Karembu, 2023
Genome Editing in Tef for Uplifting Productivity (GETUP)	Lodging resistance	MSU, EIAR, Corteva, DDPSC	CRISPR-Cas9	Deployment and Commercialization (D & C)	4.9	Donald Danforth Plant Science Center & Corteva	Beyene et al., 2022
Striga Smart Sorghum for Africa Project	Striga resistance	Kenyatta University (Lead), Addis Ababa University (Ethiopia), International Service for the Acquisition of Agri-biotech Applications ISAAA) AfriCenter	CRISPR-Cas9	R & D	3.8	Feed the Future of USAID	Iraki, Runo & Karembu, 2023

3.7. Human Capital and Institutional Capacity

3.7.1. Summary of Existing Human Capacity

Ethiopia is experiencing a rapid buildup of human capacity in modern biotechnology and genome editing (GEd), transitioning from basic research to early-stage implementation of molecular tools for agricultural advancements. Capacity is driven by a combination of specialized government research institutes, increased academic training, and international partnerships focused on CRISPR and tissue culture. Researchers and laboratory technicians have received training in molecular biology and CRISPR-Cas9 techniques, supported by regional and international partnerships. A national biosafety advisory committee comprising 15 experts, including breeders and lawyers, has been in place since 2017 to manage technical regulations for gene-edited products (Juhar& Semere, 2017). This landscape analysis identified key scientists who have undergone different training programs in genome editing. Mizan Tesfay Abraha from Tigray Agricultural Research Institute, and Tesfahun Alemu Setotaw from the Ethiopian Institute of Agricultural Research (EIAR) were *AFPBA* 2018/19 Class IV graduates, while Allo Aman Dido from BIO and Emerging Technology Institute (BETin), LetaTulu Bedada from the Ethiopian Institute of Agricultural Research (EIAR), Allo Aman Dido from BIO AND Emerging Technology Institute (BETin), and Tilahun Negassa from Addis Ababa University, Institute of Biotechnology graduated were the 2023 Class I 2023 badge. Yemisrach Melkie Abebaw from (BETin) graduated in Class II 2024.

3.7.1.1. Women, Youth and Special Interest Groups involvement in Modern Biotech and GEd

The involvement of women, youth, and special interest groups in Modern Biotechnology is increasingly prioritized to support food security and economic growth in Ethiopia. Active efforts are ongoing to integrate women and youths into the agricultural biotechnology landscape to foster an inclusive, knowledge-based bioeconomy (Jisso et al., 2022). The proportion of scientists trained in modern biotechnology and Genome Editing (GEd) among women and youth remains low but is increasing through recent targeted fellowships (2022–2026). In recent GEd-specific programs, such as the Africa Plant Breeding Academy (AfPBA) CRISPR course, Ethiopia had 10–12 molecular scientists per cycle, with intentional efforts to include female molecular biologists (Abate, 2023). Young scientists are considered crucial for driving adoption and fostering agricultural innovation, including GEd (Kitenge et al., 2025). In addition, efforts are underway to involve youths in drafting National Communication Strategies to bridge the gap between technical training and public engagement. In December 2025, Ethiopia validated a roadmap for the

Genome Editing and Bio-Innovation Hub, which is expected to outline how the country will integrate genome editing into its broader national development strategy.

3.7.2. Research, Development and Academic Institutions

A number of R&D and academic institutions in Ethiopia are engaged in various activities related to GEd technologies. Examples include:

- Ethiopian Institute of Agricultural Research (EIAR): involved in biotech and GEd R&D, GEd communication and extension services.
- Bio and Emerging Technology Institute (BETin): involved in biotech R&D, developing national capacity in biotech and emerging technologies like GEd.
- Addis Ababa University (AAU): involved in GEd R&D, education, and research training of BSc, MSc, and PhD students through curricula.
- Hawassa University: involved in training, education, and research of BSc, MSc, and PhD students through curricula.
- Addis Ababa Science and Technology University: involved in training, education, and research of BSc, MSc, and PhD students through curricula.
- Arba Minch University: involved in training, education, and research of BSc, MSc, and PhD students through curricula.
- Adama Science and Technology University: involved in training, education, and research of BSc, MSc, and PhD students through curricula.
- Gonder University: involved in training, education, and research of BSc, MSc, and PhD students through curricula.
- Haramaya University: involved in education and research training of BSc and MSc students through curricula.
- Debre Birhan University: involved in education and research training of BSc and MSc students through curricula.
- Mekelle University: involved in education and research training of BSc and MSc students through curricula.
- Walkite University: involved in education and research training of BSc and MSc students through curricula.
- Ambo University: involved in education and research training of BSc and MSc students through curricula.

- Jima University: involved in education and research training of BSc and MSc students through curricula.

3.7.3. Training and Professional Development

Specialized training in genome editing has been provided through regional and international programs, though domestic replication remains constrained by limitations in infrastructure and the regulatory framework (*see Table 5*). The Innovative Genomics Institute (IGI) offers intensive CRISPR courses for postdoctoral and mid-level research scientists, with annual four-week programs that accommodate four participants per cohort. The International Institute of Tropical Agriculture (IITA) provides comparable CRISPR training with similar duration and targeting, while AUDA-NEPAD's GEd initiative sponsors regional training programs aligned with continental capacity-building objectives (Karembu et al., 2023).

The Donald Danforth Plant Science Center offers specialized six- to twelve-month visiting scientist programs focused on genome editing in teff, with placements every six months for two mid-level research scientists. These extended placements provide deeper technical immersion and direct participation in active research projects, building sustainable capacity through experiential learning rather than short-course exposure (Beyene et al., 2022; Ngure & Karembu, 2023). A consistent constraint across all training programs is the delayed domestic replication of learned technologies, stemming from the prior absence of enabling legal frameworks. With the publication of the 2025 Guideline, this constraint is being addressed, though infrastructure gaps in equipment, reagents, and technical support continue to impede the immediate transfer of technology. Training program participants frequently report challenges in applying advanced techniques upon return due to resource limitations, suggesting a need for integrated capacity-building approaches that combine individual training with institutional infrastructure development (EPA, 2025; Karembu et al., 2023).

Table 5. Overview of Training Programs on Genome Editing

Institution / Organization	Training Program	Target Audience / of Trainees per year	Frequency	Duration	Gaps Identified
Innovative Genomic Institute	CRISPR Course	Postdoctoral, Mid-level research scientist (4 in total)	Annually	4 weeks	Delayed in replicating the technologies domestically due to a lack of a legal framework
International Institute of Tropical Agriculture (IITA)	CRISPR Course	Postdoctoral, Mid-level research scientist (4 in total)	Annually	4 weeks	Delayed in replicating the technologies domestically due to a lack of a legal framework
The Donald Danforth Plant Science Center	CRISPR Course in GEd on Tef	Visiting scientists, Mid-level research scientists (2 in total)	Every six months	6-12 months	Delayed in replicating the technologies domestically due to a lack of a legal framework
AUDA-NEPAD (GEd initiative)	CRISPR Course	Postdoctoral, Mid-level research scientist (4 in total)	Annually	4 weeks	Delayed in replicating the technologies domestically due to a lack of a legal framework

Additionally, several institutions of higher education, particularly universities, have emerged as centers of excellence in genome editing research and training, with programs spanning crop improvement and potential medical applications. Improving the quality of education and training in biotechnology is essential to generate the human capital necessary for innovation and the application of technology in national development. Ethiopia has expanded biotechnology education significantly, with more than 30 public universities now offering biotechnology courses, though the distribution of programs remains uneven between undergraduate and advanced-degree offerings (*see Table 6*). Most institutions maintain undergraduate programs, while fewer provide specialized training at MSc and PhD levels, creating a bottleneck in advanced technical capacity (Karembu et al., 2023; Spielman et al., 2013). Addis Ababa University maintains the most comprehensive portfolio through its Institute of Biotechnology, offering postgraduate programs in agricultural, industrial, and medical biotechnology, including doctoral programs in industrial and medical biotechnology (AAU, 2023). Haramaya University provides MSc-level training in agricultural biotechnology, while Hawassa University offers specialized tracks in plant and animal biotechnology, with doctoral programs in both domains. Debre Birhan University, Gonder University, Mekelle University, and Addis Ababa Science and Technology University have developed differentiated programs addressing environmental, industrial, and bioinformatics applications (Table 5). However, substantial gaps persist in training and capacity building for modern biotechnology and biosafety, particularly for emerging technologies such as genome editing. Curriculum updating lags technological advances, practical training opportunities remain limited by infrastructure constraints, and interdisciplinary programs integrating molecular biology, bioinformatics, and regulatory science are scarce (Karembu et al., 2023). The establishment of the Bio and Emerging Technology Institute (BETin) represents a government response to these challenges, though its operational capacity is still under development.

Table 6. Universities Offering Training in Biotechnology and GED-Related Courses in Ethiopia

Institution	BSc Program	Master's Program	PhD Program
Addis Ababa University	Biology	Agricultural Biotechnology	Industrial Biotechnology Medical Biotechnology
Haramaya University	Biology	Agricultural Biotechnology	No PhD Program in Biology and Agricultural Biotechnology
Debre Birhan University	General Biotechnology	Biotechnology	No PhD program in Biotechnology
Hawassa University	Biotechnology	Plant Biotechnology	Animal Biotechnology
Gonder University	General Biotechnology	Environmental & Industrial Biotechnology	Agricultural Biotechnology Medical Biotechnology
Mekelle University	Biotechnology	Bioinformatics	No PhD program in Biotechnology
Addis Ababa Science and Technology University	General Biotechnology	Plant Biotechnology Animal Biotechnology	Industrial Biotechnology
Ambo University	Biology	Biotechnology	No PhD program in Biotechnology
Jima University	Biology	Plant Biotechnology	No PhD program in Biotechnology
Adama Science and Technology University	Applied Biology	Biotechnology	Biotechnology
Walkite University	Biotechnology	Biotechnology	No PhD program in Biotechnology
Arba Minch University	Biology	Biotechnology	Biotechnology
Bahir Der University	No undergraduate course in Biotechnology	Agricultural Biotechnology Health Biotechnology Environmental Biotechnology Industrial Biotechnology	Agricultural Biotechnology Health Biotechnology Environmental Biotechnology Industrial Biotechnology

3.7.3.1. Role of Education and Training Authority (ETA)

The recognition and quality of the degrees in biotechnology and GEd-related courses offered at the universities listed in Table 6 benefit from the roles of the Education and Training Authority (ETA), formerly the Higher Education Relevance and Quality Assurance Agency (HERQA). The ETA is the overarching federal body responsible for the following:

- Issuing institutional and programmatic accreditations.
- Conducting quality audits and evaluations.
- Authenticating degrees and determining educational equivalency for qualifications.

3.8. Infrastructure and Equipment

3.8.1. Universities

Many Ethiopian universities began tissue culture work with local plants decades ago, later developing full biotechnology institutes and postgraduate training programs. Infrastructure quality varies widely among institutions. Addis Ababa University has fully equipped BSL-2 laboratories, greenhouses, and supporting facilities, but faces procurement issues, foreign currency shortages, and limited local availability of consumables and chemicals, which affect operations. Universities like Jimma, Hawassa, Gonder, Mekelle, and Addis Ababa Science and Technology have partially equipped labs with similar challenges, including procurement hurdles, forex restrictions, and reagent shortages (see Table 7). All institutions need specialized procurement systems and greater private sector involvement to tackle supply chain vulnerabilities (Karembu et al., 2023; Spielman et al., 2013).

Table 7. Assessment of Biosafety Laboratory Facilities of Universities

Institution	Biosafety Level	Status	Limitations	Support Needed
Addis Ababa University	Lab, BSL2 Greenhouse	Fully equipped,	Procurement problem, shortage of hard currency, shortage of consumables/chemicals on the local market	Specialized procurement system, private sector engagement
Jimma University	Lab, Greenhouse, field trials	Not fully equipped	Procurement problem, shortage of hard currency, shortage of consumables/chemicals on the local market	Specialized procurement system, private sector engagement

Hawassa University	Lab, Greenhouse, field trials	Not fully equipped	Procurement, shortage of hard currency, shortage of consumables/chemicals on the local market	Specialized procurement system, private sector engagement
Gonder University	Lab, Greenhouse, field trials	Not fully equipped	Procurement, shortage of hard currency, shortage of consumables/chemicals on the local market	Specialized procurement system, private sector engagement
Mekelle University	Lab, Greenhouse, field trials	Not fully equipped	Procurement, shortage of hard currency, shortage of consumables/chemicals on the local market	Specialized procurement system, private sector engagement
Addis Ababa Science and Technology University	Lab, Greenhouse, field trials	Not fully equipped	Procurement, shortage of hard currency, shortage of consumables/chemicals on the local market	Specialized procurement system, private sector engagement

3.8.2. Research Institutions

Research institutions have modest laboratory infrastructure for genome-editing projects, with capacity differences between flagship and peripheral facilities (see Table 8). The Ethiopian Institute of Agricultural Research has developed excellent laboratories with support from the World Bank through the Agricultural Research and Training Project (ARTP) and the Rural Capacity Building Project (RCBP), equipping them with essential molecular biotechnology and tissue culture facilities for genome editing. The Bio and Emerging Technology Institute (BETin), established by the government, boasts advanced facilities, including second-generation sequencing machines, to support national genomic research. Similarly, the Armauer Hansen Research Institute maintains fully equipped BSL-2 labs and greenhouse facilities to support biomedical and agricultural biotechnology research (see Table 8). Common challenges across all institutions include limitations in procurement systems, shortages of hard currency for importing equipment and reagents, and limited local access to specialized consumables and chemicals, highlighting the need for increased private-sector involvement and specialized procurement mechanisms (Karembu et al., 2023).

Table 8. Assessment of Biosafety Laboratory Facilities of Research Institutions

Institution	Biosafety Level	Status	Limitations	Support Needed
Ethiopian Institute of Agricultural Research	Lab, BSL2 Greenhouse	Fully equipped	Procurement problem, shortage of hard currency, shortage of consumables/chemicals on the local market	Specialized procurement system, private sector engagement
Bio and Emerging Technology Institute	Lab, BSL2 Greenhouse	Fully equipped	Procurement problem, shortage of hard currency, shortage of consumables/chemicals on the local market	Specialized procurement system, private sector engagement
Armauer Hansen Research Institute	Lab, BSL2 Greenhouse	Fully equipped	Procurement problem, shortage of hard currency, shortage of consumables/chemicals on the local market	Specialized procurement system, private sector engagement

3.8.2.1. Role of the Ministry of Innovation and Technology

The ministry is the primary statutory body responsible for setting science, technology, and research priorities in the country. It is mandated to develop national policies, coordinate research and innovation systems, and promote scientific advancement, including modern biotechnology and GEd, to drive socio-economic development. For specific sectors, research regulation and coordination are delegated to the following statutory and federal authorities:

- **The Ministry of Science and Higher Education (MoSHE)** coordinates academic research, sets university research agendas, and manages national ethical review frameworks.
- **The EIAR,** under the **Ministry of Agriculture (MoA),** is the statutory body that generates, develops, and coordinates agricultural technologies and research systems nationwide.

3.9. Traditional and Staple Crops, Livestock, Agroforestry, and Fisheries Varieties/ Breeds for Improvement Using GEd

Ethiopia's agricultural development planning, including the Ten-Year Development Plan (2021-2030) and successive Growth and Transformation Plans (GTPs), emphasizes productivity enhancement, modernization, and commercialization through intensive labor and land use, expanded irrigation, optimized fertilizer use, technology deployment, and infrastructure development (FDRE, 2019, 2021). Biotechnology has been explicitly identified as a critical tool for achieving these ambitious objectives, with genome editing positioned for high-impact applications in orphan crops that receive limited global research investment (Tadele, 2019; Mba et al., 2022).

The three genome-editing projects focus on crops with notable socio-economic importance and potential: sorghum, Ethiopia's fourth most crucial cereal used for food, beverages, and fodder, can expand from 4.1 to over 6 million tons with Striga resistance; Ethiopian mustard, a nascent oilseed, could grow from 75,000 to 2.3 million tons through breeding and market development; and teff, Ethiopia's staple and high-value cereal, could increase from 6.2 to 7 million tons with lodging-resistant varieties, reducing losses and aiding mechanization. All three benefit from CRISPR-Cas9 and meet market demand. However, genome editing in livestock, fisheries, marine resources, and agroforestry remains unaddressed, despite their significance, and future efforts should include animal health, productivity, and climate resilience in cattle, small ruminants, and poultry.

Table 9. Priority Target Crops and Traits in Genome Editing Research

Organism / Species	Trait improvement of Interest	Socioeconomic Justification	GEd Potential (Low/Medium/High)	Existing R&D	Actual Annual Production Capacity (tons	) Expected Annual Production Capacity (tons)
Sorghum	Resistance to the parasitic Striga weed	4th most important cereal after wheat, teff, and maize; used for food, beverages (e.g., injera, beer), and fodder	High	Conventional breeding and GEd technology	4.1 million	Over 6 million
Ethiopian mustard	Low erucic acid	Emerging oilseed with potential for local edible oil production and industrial use; niche markets for low erucic varieties	High	Conventional breeding and GEd technology	75,000	2.3 million
Teff	Lodging resistance	Cultural staple; accounts for 30% of cereal land, top value cereal	High	Conventional breeding, mutation breeding, and GEd technology	6.2 million	7 million

3.10. Intellectual Property Rights and Benefit Sharing

Ethiopia's regulatory framework for intellectual property rights (IPRs) and the benefit-sharing of genetic resources, including those used in genome editing, derives from commitments under the Nagoya Protocol and aims to ensure sovereign rights over genetic resources and equitable benefit-sharing with provider communities (FAOLEX, 2012; CBD, 2014).

The Ethiopian Intellectual Property Authority (EIPA) administers and implements state policies on intellectual property to strengthen IPR protection in the country. Key IPRs under EIPA include:

Patents and Inventions: It protects Inventions, Minor Inventions, and Industrial Designs through Proclamation No. 123/1995 and Council of Ministers Regulation No. 12/1997.

Trademarks: The Trademark Registration and Protection Regulation No. 273/2012 regulates them.

Copyright and Neighboring Rights: It is regulated through Copyright and Neighboring Rights Protection Proclamation No. 410/2004; Copyright and Neighboring Rights Protection (Amendment) Proclamation No. 872/2014, and Council of Ministers Regulation No. 305/2014.

Industrial Designs and Utility Models are regulated by Proclamation No. 123/1995.

The plant breeder's rights are, however, protected by the Plant Variety Release, Protection and Seed Quality Control Directorate, Ministry of Agriculture, Plant Breeders Rights through the Plant Breeders Rights Proclamation No. 1068/2017. A summary of these IPRs is presented in Table 10.

Table 10. A summary of Intellectual Property Rights (IPRs), their Institutions and Laws in Ethiopia

Intellectual Property Rights (IPRs)	**Organization / Institution**	**Laws (Acts) / Regulations / Protocols / Treaties**
Patents and Inventions	Ethiopian Intellectual Property Authority (EIPA)	Inventions, Minor Inventions, and Industrial Designs Proclamation No. 123/1995; Council of Ministers Regulation No. 12/1997
Trademarks		Trademark Registration and Protection Regulation No. 273/2012
Copyright and Neighboring Rights		Copyright and Neighboring Rights Protection Proclamation No. 410/2004; Copyright and Neighboring Rights Protection (Amendment) Proclamation No. 872/2014; Council of Ministers Regulation No. 305/2014

Industrial Designs and Utility Models		Proclamation No. 123/1995
Plant Breeders Rights	Plant Variety Release, Protection and Seed Quality Control Directorate, Ministry of Agriculture	Plant Breeders' Rights Proclamation No. 1068/2017

3.11. Private Sector Participation

The private sector will play a critical role in the commercialization of GEd products by providing testing sites and conducting on-farm trials of GEd-developed lines (Table 11). Private-sector seed companies will be responsible for varietal breeding, seed production, and distribution to both small- and large-scale farmers. They will also be the primary focus of seed production research before the commercialization of GEd products. Some of the private seed companies operating in Ethiopia are: Ethiopian Seed Enterprise, Biniyam Mulat Private Seed Enterprise, Green Agro Solutions PLC, MIDROC Investment Group, Luna Farms of Luna Group, and Corteva Agriscience. In addition, there are international entities such as BASF (Germany), Tropic Biosciences (UK), and 2Blades Foundation. Currently, the national agricultural research system is conducting numerous Public-Private Projects, including TELA maize, a typical PPP model involving international organizations and other NAR partners.

Table 11. Overview of Private Sector Companies Involved in Seed Production & Distribution, Crop Improvement & Biotech/Ged Activities in Ethiopia

Company	Type (Multinational, Regional, National)	GEd Activities	Partnerships	Challenges Faced	Investment Interest
Ethiopian Seed Enterprise	Government	No GEd	ATI, Ethiopian Seed Association (ESA)	Policy & regulatory bottlenecks, limited access to early generation seed (EGS),	Crop Breeding, seed production & distribution
Harvest General Trading (HGT)	National	No GEd	Agro-dealers,	Policy & regulatory bottlenecks, marketing and distribution inefficiencies	Supplies of agricultural inputs like seed
Biniyam Mulat Private Seed Enterprise	National	No GEd	Outgrower farmers, Agro-dealers	Policy & regulatory bottlenecks, limited access to early generation seed (EGS)	Crop breeding, seed production, and distribution
Green Agro Solutions PLC	National	No GEd	EIAR, CIMMYT, Netherlands Development Organization (SNV), Agro-dealers	Policy & regulatory bottlenecks, marketing and distribution inefficiencies	Supplies of agricultural inputs like seed
MIDROC Investment Group	National	No GEd	Ethio Agri-CEFT Plc (subsidiary), EIAR	Policy & regulatory bottlenecks, marketing and distribution inefficiencies	Seed production and distribution
Luna Farms	Regional	No GEd	EIAR, IFC, Luna Core	Policy & regulatory bottlenecks, marketing and distribution inefficiencies	Seed and feed production and distribution
BASF (Germany)	Multinational	No GEd	Nunhems, SNV, Solidaridad, Fair Planet, JoyTech Plc.	Policy & regulatory bottlenecks	Vegetable seed production and distribution
Corteva Agriscience	Multinational	No GEd	EIAR, AGRA, ACDI/VOCA, ILRI	Policy & regulatory bottlenecks	Seed production and distribution

3.12. Funding and Investment Landscape

Genome editing offers market growth prospects but faces investment gaps, especially in developing countries like Ethiopia. Challenges include limited funding, restricted access to proprietary tech, underrepresentation in research, and the need for ethical frameworks. Infrastructure deficits, especially in African nations, hinder progress. Ethiopia's funding comes from various sources: government support for Bt cotton, transgenic enset, and coffee (~$140,350 over five years); US aid ($3.5 million over five years); Gates Foundation ($4.9 million over five years); and Swedish aid for mustard oil quality. Private sector interest depends on regulatory clarity and product pipeline. Strategic investment could boost crop productivity, nutrition, resistance, and food security, positioning Ethiopia as a regional biotech leader (Tadele, 2019; Mba et al., 2022).

Table 12. Overview of National and Other Funding Sources for Genome Editing

Funder/Donor	Organization Type	GEd Project	Amount (USD)	Duration	Recipient Institution(s)	Area of Focus
Government funding	Government	Bt cotton, Transgenic enset, Coffee GEd	140,350	5 Years	EIAR	Indigenous crops
United States Agency for International Development (USAID)	Development partner	Striga Smart Sorghum for Africa Project (Collaborative Project)	3.5 million	5 Years	AAU	Striga-resistant Sorghum
Gates Foundation (formerly, Bill & Melinda Gates Foundation – BMGF)	Development partner	Genome Editing in Tef for Uplifting Productivity (GETUP)	4.9 million	5 Years	EIAR	Logging-resistant Tef
Swedish International Development Cooperation Agency (SIDA)	Development partner	Improving the oil qualities of Ethiopian mustard (Brassica carinata)	Funded together with other projects	4 Years	AAU	Oil quality in Brassica
Open Innovation Fund – Corteva	Private sector/ Industry	Potential investor in the future				

4. RECOMMENDATIONS AND POLICY OPTIONS

4.1 Recommendations

Based on the comprehensive landscape analysis, the following five recommendations are prioritized as critical success factors for establishing Ethiopia as Africa's leader in agricultural genome editing. These recommendations are sequenced to build upon one another, creating a coherent pathway from immediate regulatory foundations to long-term transformative impact.

i. Operationalize the 2025 Regulatory Framework Through the GET-UP Project

Regulatory certainty is the prerequisite for all genome-editing activities. Without a functional implementation of the 2025 Guideline on Regulation of Genome-Edited Products, research will remain confined to laboratories, private investment will stay on hold, and capacity building will lack practical application (Section 3.2; EPA, 2025; Entine *et al.,* 2021).

ii. Launch a National Orphan Crop Genome Editing Initiative

Ethiopia's global leadership in teff, sorghum, and finger millet confers a unique comparative advantage in crops that multinational research has largely neglected. Current projects are externally driven and fragmented, lacking strategic coordination. A national initiative would concentrate resources, build domestic scientific leadership, and address priority food security challenges (Section 3.8; Tadele, 2019; Mba et al., 2022).

iii. Build Regional Research Infrastructure and Human Capital

Infrastructure assessment reveals a severe concentration in Addis Ababa, with regional universities lacking functional capacity. Human capital gaps, including a shortage of PhD-level scientists and limited practical training, constrain the replication of technology and innovation. Without distributed infrastructure and a sustained talent pipeline, genome editing remains elite, urban, and dependent on external expertise (Karembu et al., 2023; Spielman et al., 2013).

iv. Establish Functional Seed Systems and Market Access

Without viable pathways from the laboratory to the farmer's field, genome-editing innovations fail to achieve development impact. Current seed systems are fragmented, private-sector engagement is limited, and regional market access is inadequate. GEd varieties require specific regulatory treatment, certification standards, and distribution mechanisms distinct from those for conventional and transgenic crops (Spielman et al., 2013; Zilberman et al., 2018).

v. Mobilize Sustainable Financing and Build Public Trust

Current funding is inadequate and heavily reliant on external sources ($140,350 government allocation versus $4.9 million in external project funding). Public understanding remains limited, and misinformation poses risks to social license. Without diversified, sustainable financing and broad-based public trust, genome editing programs remain fragile and politically vulnerable (Karembu et al., 2023; Adem et al., 2022; Hartley et al., 2022).

4.2 Policy Options

Ethiopia is developing its genome editing policy by setting biosafety guidelines to promote agricultural innovation, distinguishing pathways based on foreign gene presence, and encouraging responsible tech use while supporting research. The convergence of policies, scientific capacity, and agricultural challenges creates a key window for strategic action to ensure genome editing meets development goals. The policy actions address immediate capacity needs and long-term institutional growth, spanning short-term and medium-to-long-term timelines.

a. Short-Term Policy Actions (next 12–24 months)

i. Capacity Building for Regulatory Implementation: Immediate investments are needed to boost national genome editing regulation, aligning with global best practices and local needs. Priorities include training reviewers in molecular characterization, risk assessment for new traits, and case-by-case evaluation under the 2025 Guideline. Technical aid should develop standardized procedures for permit review, decision documentation, and post-release monitoring. Regional exchanges with countries like South Africa, Nigeria, and Kenya, which have advanced GEd regulation, would accelerate capacity building through peer learning.

ii. Communication Strategy Development: Strategic communication builds buy-in among officials and engages stakeholders in agricultural genome editing. It should target multiple audiences: political leaders, scientific communities, farmer organizations, and civil society. Messages should highlight Ethiopia's leadership in orphan-crop improvement, climate resilience, and safety standards, while transparently addressing concerns about corporate control, biosafety, and equitable access through evidence-based dialogue.

iii. Stakeholder Engagement Mechanisms: Sustained stakeholder engagement is vital for building consensus on genome editing and governance. Regular forums should unite researchers, regulators, private sector, farmer groups, and civil society to discuss traits, research, and regulations. Ensure meaningful involvement of women farmers, affected by production constraints, and youth, inheriting technology impacts. Engagement should go beyond Addis Ababa to regional centers, acknowledging diverse agricultural systems.

iv. Operationalizing the 2025 Guideline: The new Guideline on Regulation of Genome-Edited Products needs immediate implementation through detailed procedures, forms, review timelines, and decision criteria (EPA, 2025). This involves creating a dedicated genome-editing review track with trained staff and clear metrics. Pilot testing with initial gene-edited crop applications, like the GETUP project for confined field trial approval, will help refine procedures before wider adoption.

v. Capacity Building for Regulatory Implementation: Immediate investments are needed to boost national capacity in genome editing regulation, aligning with global best practices and local needs. Priorities include training reviewers in molecular characterization, risk assessment for new traits, and case-by-case evaluations per the 2025 Guideline. Assistance should help develop standardized procedures for permit reviews, decisions, and monitoring. Regional exchanges with countries like South Africa, Nigeria, and Kenya would enhance capacity through peer learning.

vi. Communication Strategy Development: Strategic communication is vital for securing high-level support and engaging diverse stakeholders in agricultural transformation via genome editing. The strategy should target audiences like political leaders, scientists, farmers, and civil society, addressing their specific concerns. Messaging must highlight Ethiopia's leadership in orphan-crop improvement, climate resilience, and safety standards while transparently addressing concerns through evidence-based dialogue.

Vii. Stakeholder Engagement Mechanisms: Sustained stakeholder engagement is essential for building consensus on genome-editing technology and governance. Regular forums should include researchers, regulators, private-sector, farmer groups, and civil society to discuss traits, research, and regulations. Focus on meaningful participation of women farmers, who face disproportionate challenges, and youth, who will inherit technology impacts. Engagement should extend beyond Addis Ababa to regional centers, acknowledging diverse agricultural systems and constraints.

viii. Operationalizing the 2025 Guideline: The Guideline on Regulation of Genome-Edited Products calls for quick setup of administrative procedures, application forms, review timelines, and decision criteria (EPA, 2025). This involves creating a dedicated genome-editing review track with trained staff and clear metrics. Piloting the first gene-edited crop applications, like the GETUP project for confined field trials, will help improve procedures before wider implementation.

b. Medium- and Long-Term Policy Actions (2–5 years)

Regulatory Framework Development: Ethiopia should establish science-based decision-making and regulatory frameworks for genome editing across agriculture and biomedical fields, developing specific regulations for product categories, environmental release, and post-market surveillance. Harmonization with regional frameworks, supported by AUDA-NEPAD and ABNE, will promote trade, technology transfer, shared capacity, reduce duplication, and enable economies of scale. Long-term planning must consider emerging applications like gene drives, livestock editing, and stacked modifications to maintain safety while ensuring adaptability.

Institutional Strengthening: Sustained investment is essential to strengthen institutions involved in genome editing product development and commercialization (Spielman et al., 2013; Mba et al., 2022). Research institutions need upgraded laboratories, maintenance systems, and supply chains. Regulatory agencies should establish reference labs, boost bioinformatics, and enhance inspector capacity. Universities must modernize curricula with genome editing, bioinformatics, and regulatory science, and expand postgraduate training and international exchanges. The BETin requires adequate staffing, budgeting, and political support to serve as the national coordination hub.

Innovation Ecosystem Development: Creating a supportive environment for public research and local businesses to innovate and commercialize genome-editing requires comprehensive policies (Zilberman et al., 2018; Entine et al., 2021). Intellectual property policies should balance incentivizing researchers and broad access, possibly including march-in rights for food security technologies and patent pools for CRISPR tools. University tech transfer offices need enhancement to handle licensing, agreements, and spin-off creation. Developing seed systems is vital, with policies that streamline variety release, seed certification, and expand distribution to farmers. Public procurement strategies could establish markets for genome-edited products, demonstrating government commitment and encouraging early adoption.

Strategic Partnerships and Regional Leadership: Ethiopia should strengthen stakeholder collaboration to advance genome editing, aiming to become a regional hub for orphan crop improvement and regulation. This involves hosting regional training sessions and policy dialogues, and sharing regulatory data with neighbors. Partnerships with international research centers, CGIAR, the African Orphan Crops Consortium, and advanced institutes should go beyond access to technology to include co-development that enhances Ethiopian scientific leadership. Cooperation with countries like India, China, and Brazil, with their unique regulatory approaches and biotech sectors, offers alternative models to Western partnerships

Economic Growth and Market Access: Genome editing can address challenges like rising costs and improve economic growth (Tadele, 2019; Varshney et al., 2021). Ethiopia should engage in international standards, especially within Codex Alimentarius, to influence global guidelines. Trade agreements should include biotech provisions to protect traditional markets and explore premium markets for climate-resilient crops, avoiding disruptions and promoting sustainability (Zilberman et al., 2018).

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ANNEXES

Annexure 1. List of Institutions and Resource Persons Involved in the Interview

SN	NAME	SECTOR	MINISTRY/DEPARTMENT/INSTITUTION/ ORGANIZATION	POSITION	CONTACT DETAILS
1	Dr. Allo Aman	Research Institutes	Bio and Emerging Technology Institute (BETin)	Director, Plant Biotechnology Directorate	alloaman2010@gmail.com, allo.aman@betin.gov.et
2	Mr. Messay Emana	Research Institutes	Bio and Emerging Technology Institute (BETin)	Senior Researcher	messay1979@yahoo.com
3	Mr. Obsi Deselegn	Research Institutes	Ethiopian Institute of Agricultural Research	Senior Researcher	obsibio2009@yahoo.com
4	Dr. Melaku Alemu	Research Institutes	Ethiopian Institute of Agricultural Research	Senior Researcher	melaku.alemu22@gmail.com
5	Dr. Tadessa Daba	Research Institutes	Ethiopian Institute of Agricultural Research	Lead Researcher	tadessadaba@yahoo.com
6	Dr. Dawit Tesfaye	Research Institutes	Ethiopian Institute of Agricultural Research	Senior Researcher	freedave2004@yahoo.com
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